MIL-G-23621B
22 December 1983
SUPERSEDING
MIL-G-23621A
7 December 1964

MILITARY SPECIFICATION

GUARDS, SPLASH, WHEEL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers wheel splash guards for use on motor vehicles and trailers.

1.2 Classification. Splash guards shall be one of the sizes designated in MS51331, as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 2540

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STANDARDS
FEDERAL

FED-STD-601	- Rubber: Sampling and Testing.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MS51331	- Guards, Splash, Wheel.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified, the contractor shall furnish splash guards which shall be subjected to first article inspection (see 4.4 and 6). First article inspection samples, properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent splash guards delivered to the Government shall conform to these samples in all their pertinent physical and performance attributes.

3.2 Materials. The splash guard shall be made of natural rubber, synthetic rubber, or a mixture including reclaimed rubber and reinforcing fabric. The fabric shall be good quality cotton, nylon, or equivalent material. The fabric shall be in woven layers or cords mixed homogeneously throughout the entire compound.

3.3 Design and construction.

3.3.1 Dimensions. Dimensions of the splash guard shall be as specified on Drawing MS51331.

3.3.2 Mounting. The splash guard shall have four slots or holes as shown on MS51331. The slots or holes shall be reinforced with corrosion resistant metal grommets or with a metal plate with predrilled slots or holes molded into the mounting edge.

3.3.3 Appearance. Splash guard shall be free from blisters, cracks, embedded foreign matter, or other imperfections. Both faces of the splash guard shall be rubber. All edges shall be molded and sealed. Unless otherwise specified (see 6.2), the color shall be black.

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3.4 Performance.

3.4.1 Tensile strength. The tensile strength of the material shall be not less than 1,000 pounds per square inch (psi) (see 4.6.1).

3.4.2 Elongation. The ultimate elongation shall be not less than 200 percent (see 4.6.2).

3.4.3 Ozone resistance. Rubber specimens shall not show evidence of cracking following exposure for 7 days (168 hours) to an atmosphere with ozone concentration of 50 parts ozone per 100,000,000 parts air by volume at a temperature of 40 degrees Celsius (^{o}C) (see 4.6.3).

3.4.4 Accelerated aging. After accelerated aging for a period of 16 hours in an air medium at atmospheric pressure and a temperature of 100°C $\pm$ 1°C, the material shall have a tensile strength and elongation of not less than 70 percent of the unaged values (see 4.6.4).

3.4.5 Low temperature brittleness. Specimens prepared from the basic rubber material shall not become brittle at a temperature of -40°C $\pm$ 2°C. Failure shall be defined as the occurrence of cracks, fissures, crazing, or rupture (see 4.6.5).

3.4.6 Hardness. The durometer hardness number of the material shall be not less than 60 (see 4.6.6).

3.4.7 Flexibility. At ambient temperature, when folded in the middle, and with a weight of 100 pounds applied to the fold for a period of 5 minutes, the splash guard shall not crack or show any evidence of failure (see 4.6.7).

3.5 Identification and marking. Marking shall consist of the MS number (MS51331) with appropriate dash number and the manufacturer identification in accordance with MIL-STD-130.

3.6 Workmanship. Workmanship shall be of such quality to ensure that splash guards furnished under this specification are and appear to be free from defects that would hinder their performance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection equipment. Unless otherwise specified in the contract, the supplier is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.1.2 Materials and construction. To determine conformance to 3.2, inspection and material certification records shall be maintained by the contractor. Records shall be subject to review by the Government and shall include date, part, or characteristic identification, inspection results, and disposition of lot (accepted or rejected). Corrective action taken on noted defects shall be subject to approval by the Government.

4.2 Classification of inspection.

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Control tests (see 4.5.3.2).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be conducted under the following conditions:

a. Air temperature $23^{\circ} \pm 10^{\circ}C$
b. Barometric pressure $\overline{7}25 + 50$mm HG $- 75$
c. Relative humidity 50 $\pm$ 30 percent

4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select 20 splash guards produced under the production contract (see 6.2) for first article inspection. First article samples shall be examined as specified in table I and tested as specified in table II. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply splash guards that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 First article inspection failure. Failure of any first article sample to pass specified examinations or tests shall be cause for refusal to grant first article approval until corrective action by the contractor has been approved by the Government.

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the splash guard assemblies of one type and part number, from an identifiable

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production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.2 Quality conformance examinations.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQLs) on the basis of percent defective.

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as specified in table I.

TABLE I. Classificiation of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% defective	
101	Assembly incomplete	Visual
102	Dimensions affecting interchangeability, out of tolerance	SIE 1/
103	Hardness (less than 60)	Durometer
Minor	AQL 2.5% defective	
201	Dimensions not affecting interchangeability, out of tolerance	SIE 1/
202	Identification markings, improper	Visual
203	Workmanship, faulty	Visual

1/ SIE = Standard Inspection Equipment

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TABLE II. Classification of tests.

Characteristic	Requirement	Test	First article	Control
Tensile strength	3.4.1	4.6.1	X	X
Elongation	3.4.2	4.6.2	X	X
Ozone resistance	3.4.3	4.6.3	X	
Accelerated aging	3.4.4	4.6.4	X	X
Low temperature brittleness	3.4.5	4.6.5	X	X
Hardness	3.4.6	4.6.6	X	
Flexibility	3.4.7	4.6.7	X	X

4.5.3 Quality conformance tests.

4.5.3.1 Control tests. control tests shall be conducted on one splash guard assembly from each lot of 200 units consecutively produced, except that not more than one test shall be performed in a three month period, nor less than one test in a six month period. The splash guard assemblies shall be subjected to the tests specified in table II.

4.5.3.2 Failure. Failure of any splash guard assembly to pass any of the specified inspections shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.6 Methods of inspection.

4.6.1 Tensile strength. Tensile strength shall be determined in accordance with FED-STD-601, method 4111, to verify conformance to 3.4.1. Tensile strength shall be not less than 1,000 pounds per square inch.

4.6.2 Elongation. The ultimate elongation shall be determined in accordance with FED-STD-601, method 4121, to verify conformance to 3.4.2. Elongation shall be not less than 200 percent.

4.6.3 Ozone resistance. Ozone resistance test shall be performed in accordance with FED-STD-601, method 7221, to verify conformance to 3.4.3.

4.6.4 Accelerated aging. The aging test shall be performed in accordance with FED-STD-601, method 7221, to verify conformance to 3.4.4.

4.6.5 Low temperature brittleness. Brittleness test shall be performed in accordance with FED-STD-601, method 5311, to verify conformance to 3.4.5.

4.6.6 Hardness. Durometer hardness shall be determined in accordance with FED-STD-601, method 3021, to verify conformance to 3.4.6. Hardness shall be not less than 60.

4.6.7 Flexibility. The splash guard shall be folded in the middle along its length. The weight shall be placed on a board not less than 12 inches

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long with the weight resting on the central 12 inches of the fold. This position shall be maintained for not less than 5 minutes to verify conformance to 3.4.7. Splash guards shall not crack or show any evidence of failure.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level of protection shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The splash guards acquired under this specification are intended to prevent objects from being propelled rearward by the tires of motor vehicles and trailers while these vehicles and trailers are in motion.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Size of splash guard required (see 1.2).
c. Color if not as specified (see 3.3.3).
d. First article testing, if not required (see 4.4).
e. Number of samples, if other than 20 (see 4.4).
f. Selection of applicable levels of preservation, packaging, and packing (see 5.1).

Custodians:
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Review activities:
Army - AT
DCSC-SS

User activity:
Navy - MC

Preparing activity:
Army-AT

Project No. 2540-0325

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-23621B	Guards, Splash, Wheel

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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